

Amara Okoro

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EDUCATION

Imperial College London | *MSc Computer Science (Systems & Networks)* 2018–2020

- Thesis on distributed consensus protocols for Kubernetes clusters; achieved 40% reduction in consensus latency under Byzantine fault tolerance with custom RAFT implementation
- First-class honours (3.9/4.0 GPA); recipient of the Bartle Scholarship for excellence in systems architecture

University of Lagos | *BSc Computer Science* 2014–2018

- Graduated with distinction (3.87/4.0 GPA); President of the ACM Student Chapter, grew membership from 45 to 280+ members over 2 years
- Capstone project: container orchestration platform for resource-constrained environments, deployed across 12 campus servers handling 500+ concurrent users

EXPERIENCE

Datadog | *Senior Site Reliability Engineer* 2022–Present

- Architected and deployed multi-region Kubernetes infrastructure supporting 18M+ metrics/second ingestion; reduced P99 latency by 340ms through custom scheduler and admission controller tuning
- Led incident response program across 4 SRE teams (35 people); implemented automated runbook generation reducing MTTR by 65% (from 24 min to 8.4 min) for 120+ alert types
- Designed cost optimization framework leveraging spot instances and bin-packing algorithms; saved \$4.2M annually while maintaining 99.99% SLA

Cloudflare | *DevOps Engineer II* 2020–2022

- Operated and scaled global edge network infrastructure across 200+ data centers; increased request throughput by 2.3x (from 28M to 64M req/sec) by optimizing BGP failover logic and reducing convergence time from 3.2s to 0.8s
- Implemented GitOps-based configuration management for 600+ microservices using ArgoCD; reduced deployment error rate by 94% (from 3.8% to 0.22%) and deployment time to 4 minutes
- Built comprehensive observability platform aggregating logs, traces, and metrics from 45K containers; cut debugging time by 55% and reduced false-positive alerts by 71%

Wise | *DevOps Engineer* 2019–2020

- Containerized legacy monolithic payment system serving 8M+ customers; orchestrated migration of 15 applications to Kubernetes, reducing resource footprint by 48% and server costs by \$1.8M annually
- Designed and implemented disaster recovery procedures achieving RTO of 15 minutes and RPO of 5 minutes across 3 regions; conducted quarterly drills with 100% success rate
- Developed custom Prometheus exporters and dashboards for infrastructure telemetry; improved incident detection speed by 73% with advanced anomaly detection using statistical thresholding

Shopify | *Infrastructure Engineer* 2018–2019

- Managed production Kubernetes clusters processing 2.1M transactions/day across 4,000+ nodes; implemented custom resource quotas and network policies reducing noisy neighbor impact by 82%
- Automated infrastructure provisioning using Terraform; reduced environment setup time from 6 hours to 12 minutes and eliminated 340+ manual configuration steps
- Collaborated with data platform team to optimize PostgreSQL replication; increased replica synchronization speed by 1.4x and reduced failover time from 90 seconds to 28 seconds

PROJECTS

FluxCapacity | *Go, Kubernetes, ETCD, Prometheus* 2021–Present

- Open-source predictive auto-scaling engine for Kubernetes achieving 89% resource utilization (vs. 62% with HPA); 2.8K GitHub stars and 140+ contributors from 22 countries

- Reduced scaling latency by 340ms using ML-based demand forecasting; featured in KubeCon talk attended by 1,200+ attendees with 4.8/5.0 rating

SentryNet | *Rust, eBPF, Linux kernel*

2020–2022

- Built low-overhead network performance monitoring tool using eBPF; deployed in production at 8 companies monitoring 40K+ endpoints with <2% CPU overhead vs. 8% for existing tools
- Released 12 stable versions with 1.2K GitHub stars; achieved 15K monthly downloads from crates.io with 96% positive user feedback

ChaosWarden | *Python, Docker, Kubernetes, fault injection*

2019–2020

- Developed chaos engineering framework compatible with Kubernetes and Docker Swarm; automated resilience testing for 50+ teams, identifying 340+ potential failure modes
- Reduced time-to-resilience discovery by 78% vs. manual testing; 890 GitHub stars and adopted by 3 Fortune 500 companies

EXTRACURRICULAR

KubeCon Europe | *Session Speaker & Track Chair*

2021–2024

- Delivered 4 talks on observability and chaos engineering to audiences of 800–1,200 attendees; achieved 4.7/5.0 average rating across all sessions
- Chaired 'Infrastructure at Scale' track in 2023 (12 sessions); mentored 25 emerging speakers and increased diversity representation by 34%

Cloud Native Computing Foundation | *Maintainer, Kubernetes SIG-Reliability*

2020–Present

- Lead maintainer for reliability best practices documentation; contributed 180+ code reviews and merged 92 proposals from global community
- Organized bi-weekly office hours attracting 40–60 engineers; built mentorship program that trained 18 junior SREs for senior roles at major tech companies

TECHNICAL SKILLS

Languages – Go, Python, Rust, Bash, SQL, JavaScript

Technologies – Kubernetes, Docker, Prometheus, Grafana, ELK Stack, PostgreSQL, Redis, RabbitMQ, gRPC, Envoy, Terraform, ArgoCD, Jenkins, GitLab CI/CD

Tools – AWS (EC2, EKS, RDS, S3, Lambda), GCP (GKE, Cloud Run, Bigtable), Azure (AKS, Cosmos DB), Datadog, New Relic, HashiCorp suite, Vault, Consul, Linux, Git